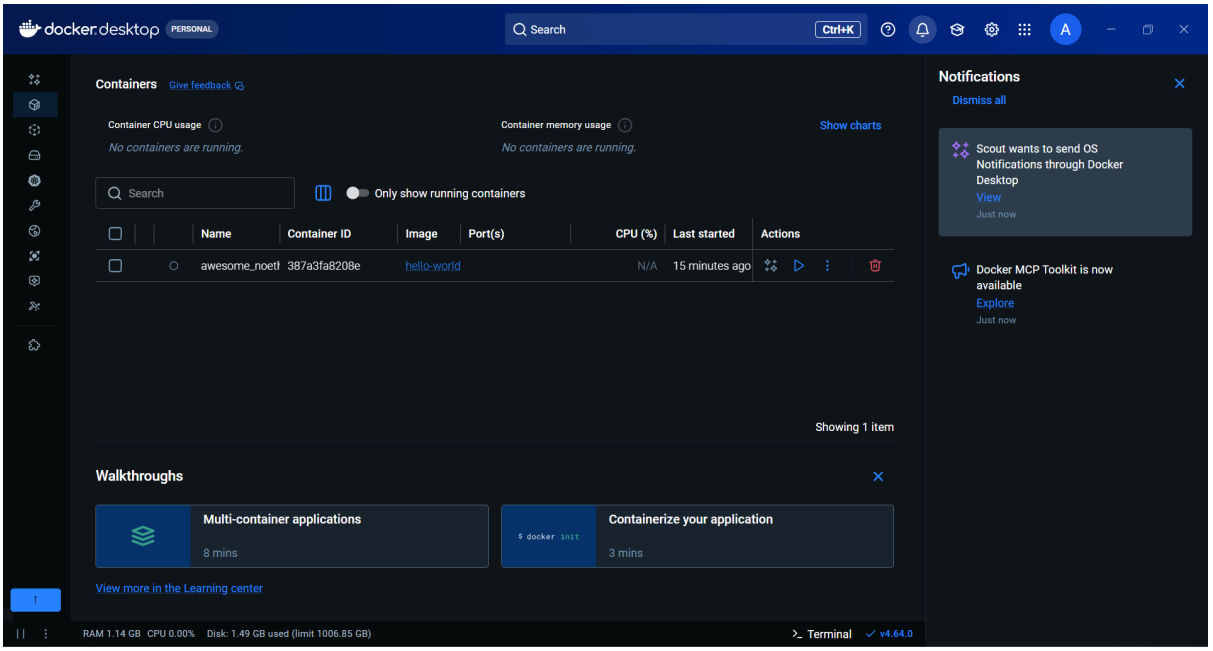
WSL Installed

```
Windows PowerShell
PS C:\Users\abina> wsl --list --verbose
  NAME              STATE           VERSION
* Ubuntu-22.04      Running         2
  docker-desktop    Running         2
PS C:\Users\abina>
```

Docker installed

```
ubuntu@LAPTOP-NAK4ILBA:~$ sudo service docker start
docker ps
Failed to start docker.service: Unit docker.service not found.
-bash: /mnt/c/Program Files/Docker/Docker/resources/bin/docker: No such file or directory
ubuntu@LAPTOP-NAK4ILBA:~$ docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES
ubuntu@LAPTOP-NAK4ILBA:~$ docker run hello-world
Unable to find image 'hello-world:latest' locally
latest: Pulling from library/hello-world
17eec7bbc9d7: Pull complete
ea52d2000f90: Download complete
Digest: sha256:85404b3c53951c3ff5d40de0972b1bb21fafa2e8daa235355baf44f33db9dbdd
Status: Downloaded newer image for hello-world:latest

Hello from Docker!
This message shows that your installation appears to be working correctly.

To generate this message, Docker took the following steps:
 1. The Docker client contacted the Docker daemon.
 2. The Docker daemon pulled the "hello-world" image from the Docker Hub.
    (amd64)
 3. The Docker daemon created a new container from that image which runs the
    executable that produces the output you are currently reading.
 4. The Docker daemon streamed that output to the Docker client, which sent it
    to your terminal.

To try something more ambitious, you can run an Ubuntu container with:
 $ docker run -it ubuntu bash

Share images, automate workflows, and more with a free Docker ID:
 https://hub.docker.com/

For more examples and ideas, visit:
 https://docs.docker.com/get-started/

ubuntu@LAPTOP-NAK4ILBA:~$
```



Installed Kubectl

```
ubuntu@LAPTOP-NAK4ILBA:~$ sudo apt update
sudo apt install -y curl
[sudo] password for ubuntu:
Hit:1 http://security.ubuntu.com/ubuntu jammy-security InRelease
Hit:2 http://archive.ubuntu.com/ubuntu jammy InRelease
Hit:3 http://archive.ubuntu.com/ubuntu jammy-updates InRelease
Hit:4 http://archive.ubuntu.com/ubuntu jammy-backports InRelease
Reading package lists... Done
E: The method driver /usr/lib/apt/methods/[download.docker.com](https could not be found.
N: Is the package apt-transport-[download.docker.com](https installed?
E: Failed to fetch [download.docker.com](https://download.docker.com/linux/ubuntu)/dists/jammy/InRelease
E: Some index files failed to download. They have been ignored, or old ones used instead.
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
curl is already the newest version (7.81.0-1ubuntu1.23).
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
ubuntu@LAPTOP-NAK4ILBA:~$ curl -LO https://storage.googleapis.com/kubernetes-release/release/$(curl -s https://storage.googleapis.com/kubernetes-release/rel
ease/stable.txt)/bin/linux/amd64/kubectl
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left  Speed
100 53.7M  100 53.7M    0     0  3832k      0  0:00:14  0:00:14 --:--:-- 7163k
ubuntu@LAPTOP-NAK4ILBA:~$ chmod +x kubectl
ubuntu@LAPTOP-NAK4ILBA:~$ sudo mv kubectl /usr/local/bin/
ubuntu@LAPTOP-NAK4ILBA:~$ kubectl version --client
Client Version: v1.31.0
Kustomize Version: v5.4.2
ubuntu@LAPTOP-NAK4ILBA:~$
```

Installed minikube

```
ubuntu@LAPTOP-NAK4ILBA:~$ curl -LO https://storage.googleapis.com/kubernetes-release/release/$(curl -s https://storage.googleapis.com/kubernetes-release/rel
ease/stable.txt)/bin/linux/amd64/kubectl
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left  Speed
100 53.7M  100 53.7M    0     0  3832k      0  0:00:14  0:00:14 --:--:-- 7163k
ubuntu@LAPTOP-NAK4ILBA:~$ chmod +x kubectl
ubuntu@LAPTOP-NAK4ILBA:~$ sudo mv kubectl /usr/local/bin/
ubuntu@LAPTOP-NAK4ILBA:~$ kubectl version --client
Client Version: v1.31.0
Kustomize Version: v5.4.2
ubuntu@LAPTOP-NAK4ILBA:~$ curl -LO https://storage.googleapis.com/minikube/releases/latest/minikube-linux-amd64
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left  Speed
100  128M  100  128M    0     0  6654k      0  0:00:19  0:00:19 --:--:-- 9499k
ubuntu@LAPTOP-NAK4ILBA:~$ sudo install minikube-linux-amd64 /usr/local/bin/minikube
ubuntu@LAPTOP-NAK4ILBA:~$ minikube version
minikube version: v1.38.1
commit: c93a4cb9311efc66b90d33ea03f75f2c4120e9b0
ubuntu@LAPTOP-NAK4ILBA:~$
```

Oracle VM VirtualBox + Extension Pack is installed set up.

```
PS C:\Users\abina> & "C:\Program Files\Oracle\VirtualBox\VBoxManage.exe" list extpacks
Extension Packs: 1
Pack no. 0:   Oracle VirtualBox Extension Pack
Version:        7.2.6
Revision:       172322
Edition:
Description:    Oracle Cloud Infrastructure integration, PXE ROM.
VRDE Module:
Crypto Module:
Usable:         true
Why unusable:
PS C:\Users\abina>
```

Started kubctl & namespaces

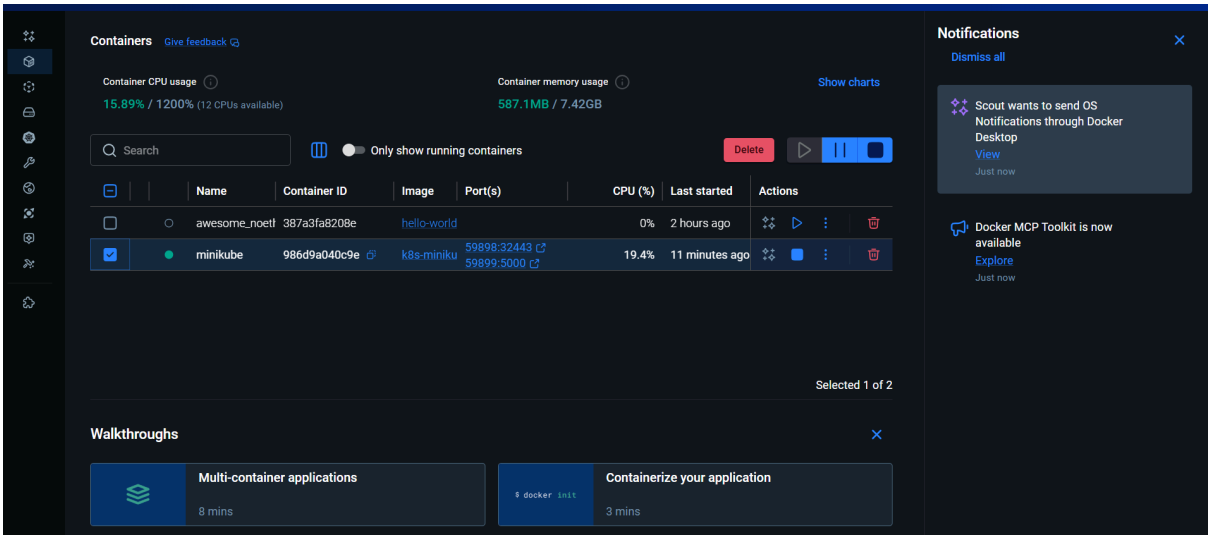
```
ubuntu@LAPTOP-NAK4ILBA:~$ minikube start --driver=docker
😄  minikube v1.38.1 on Ubuntu 22.04 (kvm/amd64)
✨  Using the docker driver based on user configuration
❗  Starting v1.39.0, minikube will default to "containerd" container runtime. See #21973 for more info.
📌  Using Docker driver with root privileges
❗  For an improved experience it's recommended to use Docker Engine instead of Docker Desktop.
Docker Engine installation instructions: https://docs.docker.com/engine/install/#server
👍  Starting "minikube" primary control-plane node in "minikube" cluster
🚜  Pulling base image v0.0.50 ...
💾  Downloading Kubernetes v1.35.1 preload ...
    > preloaded-images-k8s-v18-v1...:  272.45 MiB / 272.45 MiB  100.00% 5.39 Mi
    > gcr.io/k8s-minikube/kicbase...:  519.58 MiB / 519.58 MiB  100.00% 6.01 Mi
🔥  Creating docker container (CPUs=2, Memory=3072MB) ...
🐳  Preparing Kubernetes v1.35.1 on Docker 29.2.1 ...
🔗  Configuring bridge CNI (Container Networking Interface) ...
🔎  Verifying Kubernetes components...
    ▪ Using image gcr.io/k8s-minikube/storage-provisioner:v5
🌟  Enabled addons: storage-provisioner, default-storageclass

❗  /usr/local/bin/kubectl is version 1.31.0, which may have incompatibilities with Kubernetes 1.35.1.
    ▪ Want kubectl v1.35.1? Try 'minikube kubectl -- get pods -A'
🏄  Done! kubectl is now configured to use "minikube" cluster and "default" namespace by default
ubuntu@LAPTOP-NAK4ILBA:~$ kubectl get nodes
NAME       STATUS   ROLES           AGE   VERSION
minikube   Ready    control-plane   18s   v1.35.1
ubuntu@LAPTOP-NAK4ILBA:~$ kubectl get namespaces
NAME              STATUS   AGE
default           Active   34s
kube-node-lease   Active   34s
kube-public       Active   34s
kube-system       Active   34s
ubuntu@LAPTOP-NAK4ILBA:~$
```

Name space created

```
ubuntu@LAPTOP-NAK4ILBA:~$ kubectl create namespace dev
namespace/dev created
ubuntu@LAPTOP-NAK4ILBA:~$ kubectl get namespaces
NAME              STATUS   AGE
default           Active   6m57s
dev               Active   21s
kube-node-lease   Active   6m57s
kube-public       Active   6m57s
kube-system       Active   6m57s
ubuntu@LAPTOP-NAK4ILBA:~$
```

Runed a pod inside namespace

```
ubuntu@LAPTOP-NAK4ILBA:~$ kubectl get namespaces
NAME              STATUS   AGE
default           Active   6m57s
dev               Active   21s
kube-node-lease   Active   6m57s
kube-public       Active   6m57s
kube-system       Active   6m57s
ubuntu@LAPTOP-NAK4ILBA:~$ kubectl run nginx --image=nginx -n dev
pod/nginx created
ubuntu@LAPTOP-NAK4ILBA:~$ kubectl get pods -n dev
NAME    READY   STATUS              RESTARTS   AGE
nginx   0/1     ContainerCreating   0          13s
ubuntu@LAPTOP-NAK4ILBA:~$
```


Containers
Give feedback
Container CPU usage
15.89% / 1200% (12 CPUs available)
Container memory usage
587.1MB / 7.42GB
Show charts
Search
Only show running containers
Delete
Name
Container ID
Image
Port(s)
CPU (%)
Last started
Actions
awesome_noetl
387a3fa8208e
hello-world
0%
2 hours ago
minikube
986d9a040c9e
k8s-miniku
59898:32443
59899:5000
19.4%
11 minutes ago
Selected 1 of 2
Walkthroughs
Multi-container applications
8 mins
$ docker init
Containerize your application
3 mins
Notifications
Dismiss all
Scout wants to send OS Notifications through Docker Desktop
View
Just now
Docker MCP Toolkit is now available
Explore
Just now